

29/7/26

Signal Processing

◦ Recap: Motivation

◦ Examples:

◦ Topics

- Discrete time signals and systems

- DTFT, Z-transform, DFT

- Sampling

- Transform analysis of LTI systems

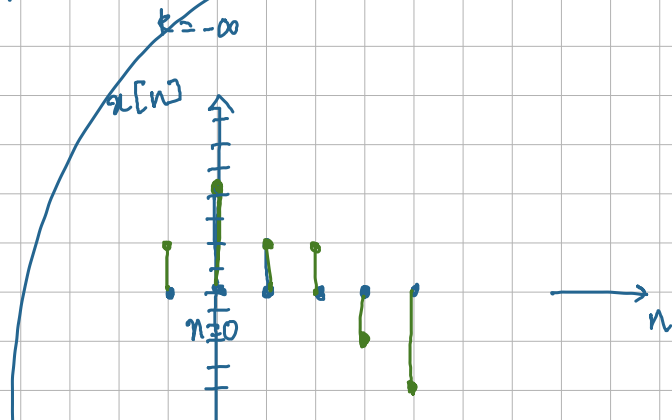
- FIR and IIR filters

- Wavelets (?)

$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$

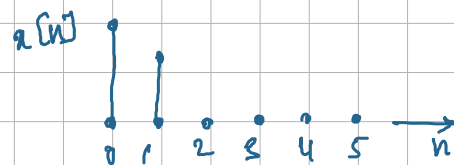
$n \in \mathbb{Z}$ } → Delta functions
①

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k] \quad - (2) \rightarrow \text{Arbitrary discrete time signal}$$



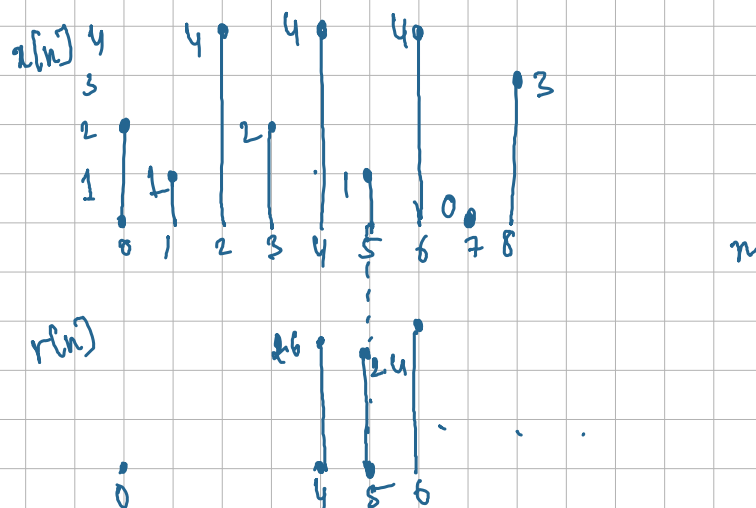
◦ 1 over in a cricket match.

$$x[0] = 6, \quad x[1] = 4, \quad x[2] = x[3] = x[4] = x[5] = 0 \quad - (3)$$



- Compute the moving average of the runs scored over the past five runs in a match.

$$r[n] = \frac{1}{5} \sum_{k=n-4}^n x[k] = \frac{1}{5} \sum_{k=0}^4 x[n-k] \quad (4)$$



$$h[n] = x[n] - x[n-1] \quad (5)$$

Derivation: $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (6)$$

